

Math 421

Friday, October 17

Announcements

- No homework due today

Theorem 11.2

Thm. Let (s_n) be a sequence.

1. If t is in $\mathbb{R}$, then there is a subsequence of (s_n) converging to t if and only if the set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is infinite for all $\epsilon > 0$.
2. If the sequence (s_n) is unbounded above, it has a subsequence with limit $+\infty$.
3. If the sequence (s_n) is unbounded below, it has a subsequence with limit $-\infty$.

In all cases, the subsequence can be taken to be monotonic.

Discussion Questions

- ❖ What is part 1 of the theorem saying you your own words?
 - If t is in $\mathbb{R}$, then there is a subsequence of (s_n) converging to t if and only if the set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is infinite for all $\epsilon > 0$.
- ❖ Convince yourselves the forward direction of part 1 is true: If there is a subsequence of (s_n) converging to t then set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is infinite for all $\epsilon > 0$.
- ❖ Convince yourselves parts 2 and 3 are true.

Theorem 11.2 – Proof of 1

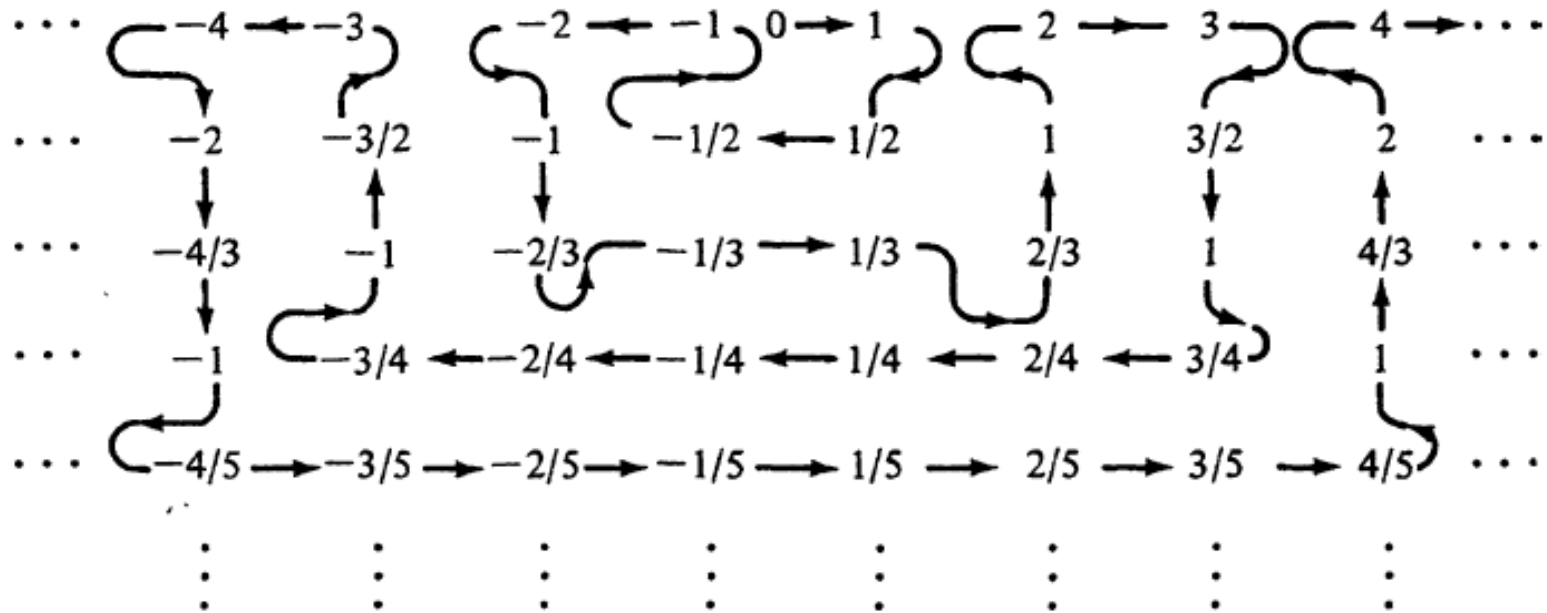
Thm. Let (s_n) be a sequence. If t is in $\mathbb{R}$, then there is a subsequence of (s_n) converging to t if and only if the set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is infinite for all $\epsilon > 0$.

Discussion.

Theorem 11.2 – Proof of 1

Discussion. (cont.)

Example – Sequence of rational numbers



Theorems involving subsequences

Theorem 11.3: If the sequence (s_n) converges, then every subsequence converges to the same limit.

Theorem 11.4: Every sequence (s_n) has a monotonic subsequence.

Discussion Question: Convince yourselves these statements are true. It could be helpful to play around with some specific examples.

Bolzano-Weierstrass Theorem

Thm: Every bounded sequence has a convergent subsequence.

Discussion Question: Why is this theorem true?

Subsequential Limit

Def. Let (s_n) be a sequence in $\mathbb{R}$. A **subsequential limit** is any real number or symbol $+\infty$ or $-\infty$ that is the limit of some subsequence of (s_n) .

Example.

Activity

For each of the following sequences, determine its set of subsequential limits.

1. (s_n) where $s_n = \frac{1}{n}$.
2. (a_n) where $a_n = n^2(-1)^n$.
3. (b_n) where $b_n = \sin\left(\frac{n\pi}{3}\right)$.
4. (r_n) a list of all rational numbers.
5. (c_n) where $c_n = n(1 + (-1)^n)$.